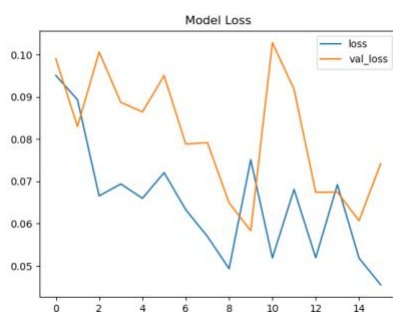


Exercise 2.5: Visual Applications of Machine Learning

Part 1: Handwriting Recognition

Actual Handwritten	Results
0	0
1	1
2	2
3	3
4	4
5	3
6	8
7	2
8	8
9	9

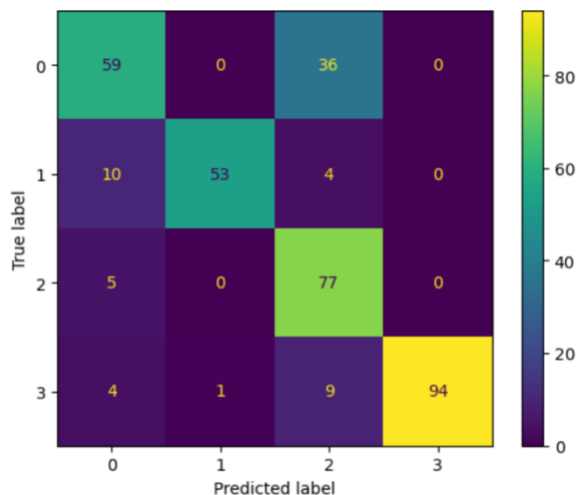
The model's accuracy is 70% because this model recognized correctly 7 out of 10. It failed to recognize number 5, 6 and 7 correctly.

Part 2: Radar Recognition

Using 16 epochs, the results are the following:

Accuracy	0.84375
Val_accuracy	0.816964
Loss	0.045502
Val_Loss	0.074189

The model's training performance shows positive progress overall. From the **loss graph**, we can see that the training loss gradually decreases, meaning the model is learning to minimize errors on the training set. However, the **validation loss** fluctuates more and occasionally spikes, indicating that the model is not consistently performing well on unseen data — a sign of slight **overfitting**. In the **accuracy graph**, the training accuracy steadily increases and remains high (around 85%), which confirms the model is fitting the training data well. The validation accuracy also improves but shows more irregular jumps, meaning the model's ability to generalize is somewhat unstable.



The confusion matrix reveals that the model has difficulty to identifying "cloudy" images accurately, often confusing them with "shine" and occasionally with "rain." Meanwhile, it performs well in recognizing "shine" and "sunrise" images, with very few errors. This indicates that the model is particularly challenged by the visual similarities between cloudy conditions and other weather types, especially "shine."

Make a Proposal for the use of GANs in weather prediction:

1. GANs can be trained to generate short-term future weather sequences (like the next few hours of radar maps), learning from historical image sequences to predict how systems like storm fronts evolve over time.
2. GANs can reconstruct missing or corrupted weather data (e.g., due to sensor failure or satellite blind spots), improving the continuity and accuracy of real-time forecasting.
3. GANs could be used to analyze satellite or radar images and automatically identify and label weather patterns such as storms, hurricanes, or cloud formations, making it easier to detect and track severe weather events.

There are several studies in this field. For instance, these are three that I found:

1. <https://www.semanticscholar.org/paper/DeepClimGAN%3A-A-High-Resolution-Climate-Data-Puchko-Link/86b9db65ff46e8dfe0ee6e3fa8b5f16f5eee0735>
2. <https://arxiv.org/pdf/2210.14266v1>
3. <https://proceedings.mlr.press/v151/price22a/price22a.pdf>.